

In [1]:

```
import pandas as pd
df = pd.read_csv('customer_shopping_behavior.csv')
```

In [2]:

```
# to see headlines
```

In [3]:

```
df.head(1)
```

Out[3]:

	Customer ID	Age	Gender	Item Purchased	Category	Purchase Amount (USD)	Location	Size	Color	Season	Review Rating	Subscription Status
0	1	55	Male	Blouse	Clothing	53	Kentucky	L	Gray	Winter	3.1	Unsubscribed

In [4]:

```
# to get info the table
```

In [5]:

```
df.info()
```

```
<class 'pandas.DataFrame'>
RangeIndex: 3900 entries, 0 to 3899
Data columns (total 18 columns):
#   Column                                Non-Null Count  Dtype  
---  -
0   Customer ID                          3900 non-null   int64  
1   Age                                   3900 non-null   int64  
2   Gender                               3900 non-null   str     
3   Item Purchased                       3900 non-null   str     
4   Category                             3900 non-null   str     
5   Purchase Amount (USD)                3900 non-null   int64  
6   Location                             3900 non-null   str     
7   Size                                  3900 non-null   str     
8   Color                                 3900 non-null   str     
9   Season                               3900 non-null   str     
10  Review Rating                         3863 non-null   float64 
11  Subscription Status                  3900 non-null   str     
12  Shipping Type                       3900 non-null   str     
13  Discount Applied                    3900 non-null   str     
14  Promo Code Used                     3900 non-null   str     
15  Previous Purchases                  3900 non-null   int64  
16  Payment Method                      3900 non-null   str     
17  Frequency of Purchases              3900 non-null   str     
dtypes: float64(1), int64(4), str(13)
memory usage: 548.6 KB
```

In [6]:

```
# to describe all about the table
```

In [7]:

```
df.describe(include = 'all')
```

Out[7]:

	Customer ID	Age	Gender	Item Purchased	Category	Purchase Amount (USD)	Location	Size	Color
count	3900.000000	3900.000000	3900	3900	3900	3900.000000	3900	3900	3900
unique	NaN	NaN	2	25	4	NaN	50	4	25
top	NaN	NaN	Male	Blouse	Clothing	NaN	Montana	M	Olive
freq	NaN	NaN	2652	171	1737	NaN	96	1755	177
mean	1950.500000	44.068462	NaN	NaN	NaN	59.764359	NaN	NaN	NaN
std	1125.977353	15.207589	NaN	NaN	NaN	23.685392	NaN	NaN	NaN
min	1.000000	18.000000	NaN	NaN	NaN	20.000000	NaN	NaN	NaN
25%	975.750000	31.000000	NaN	NaN	NaN	39.000000	NaN	NaN	NaN
50%	1950.500000	44.000000	NaN	NaN	NaN	60.000000	NaN	NaN	NaN
75%	2925.250000	57.000000	NaN	NaN	NaN	81.000000	NaN	NaN	NaN
max	3900.000000	70.000000	NaN	NaN	NaN	100.000000	NaN	NaN	NaN

In [8]:

```
# to find the null values
```

In [9]:

```
df.isnull().sum()
```

Out[9]:

```
Customer ID          0
Age                  0
Gender               0
Item Purchased       0
Category             0
Purchase Amount (USD) 0
Location            0
Size                0
Color               0
Season              0
Review Rating       37
Subscription Status  0
Shipping Type        0
Discount Applied     0
Promo Code Used      0
Previous Purchases   0
Payment Method       0
Frequency of Purchases 0
dtype: int64
```

In [10]:

```
# to fill the avg median value in the null cells
```

In [11]:

```
df['Review Rating'] = df.groupby('Category')['Review Rating'].transform(lambda x: x.fill
```

In [12]:

```
# to find again the null value
```

```
In [13]:
```

```
df.isnull().sum()
```

```
Out[13]:
```

```
Customer ID      0
Age              0
Gender           0
Item Purchased   0
Category         0
Purchase Amount (USD)  0
Location         0
Size            0
Color           0
Season          0
Review Rating    0
Subscription Status  0
Shipping Type    0
Discount Applied 0
Promo Code Used  0
Previous Purchases 0
Payment Method   0
Frequency of Purchases 0
dtype: int64
```

```
In [14]:
```

```
# to see the columns values
```

```
In [15]:
```

```
df.columns
```

```
Out[15]:
```

```
Index(['Customer ID', 'Age', 'Gender', 'Item Purchased', 'Category',
      'Purchase Amount (USD)', 'Location', 'Size', 'Color', 'Season',
      'Review Rating', 'Subscription Status', 'Shipping Type',
      'Discount Applied', 'Promo Code Used', 'Previous Purchases',
      'Payment Method', 'Frequency of Purchases'],
      dtype='str')
```

```
In [16]:
```

```
# to replace the space to _ and set the alphabet into lower case and rename the column
```

```
In [17]:
```

```
df.columns = df.columns.str.lower()
df.columns = df.columns.str.replace(' ', '_')
df = df.rename(columns={'purchase_amount_(usd)': 'purchase_amount'})
```

```
In [18]:
```

```
df.columns
```

```
Out[18]:
```

```
Index(['customer_id', 'age', 'gender', 'item_purchased', 'category',
      'purchase_amount', 'location', 'size', 'color', 'season',
      'review_rating', 'subscription_status', 'shipping_type',
      'discount_applied', 'promo_code_used', 'previous_purchases',
      'payment_method', 'frequency_of_purchases'],
      dtype='str')
```

In [19]:

```
# to create a column age_group
```

In [20]:

```
labels = ['Young Adult','Adult','Middle_aged','Senior']  
df['age_group'] = pd.qcut(df['age'],q=4, labels = labels)
```

In [21]:

```
df.columns
```

Out[21]:

```
Index(['customer_id', 'age', 'gender', 'item_purchased', 'category',  
      'purchase_amount', 'location', 'size', 'color', 'season',  
      'review_rating', 'subscription_status', 'shipping_type',  
      'discount_applied', 'promo_code_used', 'previous_purchases',  
      'payment_method', 'frequency_of_purchases', 'age_group'],  
      dtype='str')
```

In [22]:

```
# to find a specific columns
```

In [23]:

```
df[['age','age_group']].head(11)
```

Out[23]:

	age	age_group
0	55	Middle_aged
1	19	Young Adult
2	50	Middle_aged
3	21	Young Adult
4	45	Middle_aged
5	46	Middle_aged
6	63	Senior
7	27	Young Adult
8	26	Young Adult
9	57	Middle_aged
10	53	Middle_aged

In [24]:

```
# to find some specific columns
```

In [25]:

```
df[['gender','location','size']].head(5)
```

Out[25]:

	gender	location	size
0	Male	Kentucky	L
1	Male	Maine	L

	gender	location	size
2	Male	Massachusetts	S
3	Male	Rhode Island	M
4	Male	Oregon	M

In [26]:

```
# create column purchase_frequency_days
```

In [37]:

```
frequency_mapping = {
    'Fortnightly': 14,
    'Weekly': 7,
    'Monthly': 30,
    'Quarterly': 90,
    'Bi-Weekly': 14,
    'Annually': 365,
    'Every 3 Months': 90
}
df['purchase_frequency_days'] = df['frequency_of_purchases'].map(frequency_mapping)
```

In [40]:

```
df[['purchase_frequency_days', 'frequency_of_purchases']].head(9)
```

Out[40]:

	purchase_frequency_days	frequency_of_purchases
0	14	Fortnightly
1	14	Fortnightly
2	7	Weekly
3	7	Weekly
4	365	Annually
5	7	Weekly
6	90	Quarterly
7	7	Weekly
8	365	Annually

In [41]:

```
df[['discount_applied', 'promo_code_used']].head(9)
```

Out[41]:

	discount_applied	promo_code_used
0	Yes	Yes
1	Yes	Yes
2	Yes	Yes
3	Yes	Yes
4	Yes	Yes

	discount_applied	promo_code_used
5	Yes	Yes
6	Yes	Yes
7	Yes	Yes
8	Yes	Yes

In [42]:

```
# to find the when the promo used discount applied is true
```

In [43]:

```
(df['discount_applied'] == df['promo_code_used']).all()
```

Out[43]:

np.True_

In [44]:

```
# to drop column
```

In [45]:

```
df = df.drop('promo_code_used', axis = 1)
```

In [46]:

```
df.columns
```

Out[46]:

```
Index(['customer_id', 'age', 'gender', 'item_purchased', 'category',
      'purchase_amount', 'location', 'size', 'color', 'season',
      'review_rating', 'subscription_status', 'shipping_type',
      'discount_applied', 'previous_purchases', 'payment_method',
      'frequency_of_purchases', 'age_group', 'purchase_frequency_days'],
      dtype='str')
```

In [47]:

```
pip install psycopg2-binary sqlalchemy
```

Requirement already satisfied: psycopg2-binary in .\AppData\Local\Python\pythoncore-3.14-64\Lib\site-packages (2.9.12)

Requirement already satisfied: sqlalchemy in .\AppData\Local\Python\pythoncore-3.14-64\Lib\site-packages (2.0.50)

Requirement already satisfied: greenlet>=1 in .\AppData\Local\Python\pythoncore-3.14-64\Lib\site-packages (from sqlalchemy) (3.5.1)

Requirement already satisfied: typing-extensions>=4.6.0 in .\AppData\Local\Python\pythoncore-3.14-64\Lib\site-packages (from sqlalchemy) (4.15.0)

Note: you may need to restart the kernel to use updated packages.

In [49]:

```
df.to_csv(r"C:\Users\ADMIN\OneDrive\Desktop\sql\customer behavior\clean_customer_behavior")
```

In []: